

Nonlinear Dynamic Inversion based Motion planning of a Floating Satellite Platform

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Abstract—A nonlinear dynamic inversion based motion planning for levitating robotic satellite emulation platform is presented in this article. The friction-less motion of such a levitating platform has dynamic equivalency with a satellite's realistic movement in space, thus evolving as an efficient demonstration tool as a hardware-in-loop satellite motion simulator. A brief design description of the emulation platform unified with various sensor and actuator units is presented, and an associated nonlinear mathematical model representing its coupled dynamic motion is described. The proposed control design architecture manipulates the associated nonlinear dynamics, which appear due to the coupling of translational and rotational movement of the robotic platform, thereby establishing an efficient trajectory tracking capability for various operating regions without the additional requirement of gain scheduling. In order to realize the control requirement through a set of ON-OFF thruster based actuation unit, which closely resembles the reaction control system of realistic satellite design, an optimal thrust allocation logic is proposed that accounts for various physical limitations of thrusters such as thrusters as its magnitude constraint and minimum on time. The efficacy of the proposed modeling and control scheme is experimentally validated in different realistic motion planning scenarios such as point-to-point manoeuvre with terminal objectives and a superior path following capability. The experimental studies are supported with related results.

Index Terms—Nonlinear Feedback linearization Control, Floating planar platform, Friction-less platform, Nonlinear modeling.

I. INTRODUCTION

The evolution of small satellite technologies brings up the platform for the space robotics community as a fundamental enabler towards the new generation of space activity. The recent space missions, which are being scheduled or in development, target autonomous proximity operations such as rendezvous and docking, on-orbit servicing, manufacturing and assembly, active debris removal, etc. Moreover, with the advancement in micro, nano and cub sat technologies [1], next generation space operations are looking forward to integrating multiple satellite formations to collaboratively perform these complex operations collectively [2], [3] while enabling resiliency, mission endurance, and cost-effective solutions. Such emerging yet highly challenging space activities demands advanced reliable execution of highly accurate

autonomy that will invade the state of the art guidance, navigation and control (GNC) approaches [4].

However, due to uncompromising cost to failure, it is necessary to evaluate the functionality and efficacy of such advanced GNC technology prior to launching the satellite system into space [5]. Hence, high fidelity hardware in the loop emulation platform is an essential prerequisite. In view of that, several researchers have proposed various unprecedented experimental setups where space like motion and weightlessness in the microgravity environment can be imitated on Earth. Such innovative approaches includes parabolic flights [6], drop-towers [7], underwater submerging and so on. However, the planar air-bearing based levitating robotic platform [8], [9] provides the most convenient and flexible physical equivalency in terms of space like friction-less orbital motion planning.

In this context, the floating robotic platforms are usually supported with air-bearings through which compressed air is uniformly released, eventually creating an air cushion. The functional surface of air-bearings is porous in nature. This air-cushion over an extremely smooth flat surface supports the friction-less, micro-gravity orbital motion [10]–[13] to emulate space-like conditions. Such robotic platforms are equipped with multiple air realising thrusters-based actuation units that operate in On-OFF mode, equivalent to a typical spacecraft Reaction control system (RCS). It enables the robotic platform to perform planar manoeuvres in a controlled manner. The planar manoeuvrability and the thruster-based RCS actuation unit are preferred as space emulation testbed facilities. To this end, it's worth noting that there are no commercially accessible platforms for these friction-free micro gravity emulation systems to the best of our knowledge. Various government agencies and university laboratories around the globe have built their testbed facilities. Following the trend, the Robotics and AI team developed a Slider, floating platform with a compressed air tank, air bearings, an on-board flight controller, and related sensors and actuator units. The details of the mechanical design is presented in [14].

In order to establish the autonomous manoeuvrability of such robotic platforms for emulating satellite's motion, it is necessary to establish its dynamic model and corresponding motion control system. Towards this direction various progressive controller design approaches for the floating robotic platform are proposed in literature, which include conventional proportional and derivative (PD) control [15], potential field gradient [16], Linear Quadratic Regulator (LQR) [17] etc. However, most of these approaches con-

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sider simplified dynamics of the robotic platform, which does not explicitly account for the nonlinearity due to the translational and rotational motion coupling. Hence it leads to gain scheduling [18] based design tuning for the various operating region. In view of that, considering the nonlinearity of the system dynamics, a composite control architecture is developed in [19], where the feedback linearization approach is adapted to formulate an equivalent linear system dynamics which is controlled using the LQR approach.

In line with the presented background, this article aims in presenting a nonlinear dynamic inversion based closed-form, state feedback, motion control architecture, which incorporates the non-linearity and coupled motion of the platform dynamics while eliminating the requirement for tedious gain scheduling. The dynamic inversion based control design approach has successfully been applied in various application areas [20], [21]. However, typically dynamic inversion based controller generates the control requirement as a continuous function of time. In order to translate the controller command to an equivalent actuation unit, a separate control allocation logic is inevitable. Depending on the actuator type and associated architectural alignment, various control allocation and actuation logic has been considered in the literature. For example, an independent set of fixed thrusters are adapted for translational and rotational manoeuvre in [10], a set of gimbaled thruster configurations is presented in [19]. In contrast, a blended actuation mechanism consisting of a set of pulsating thrusters for translation and reaction wheel-based continuous actuation for rotational motion is incorporated in [15]. In view of that, the proposed platform design considers a more generic allocation unit consisting of eight pulsating thrusters symmetrically attached around the platform, each of which is capable of producing translation and rotational motion, typically resembles with small satellite configuration.

Thus, the main contributions of the article stem from: a) the introduction of a novel design of a planar floating platform, Slider, and establishing a nonlinear mathematical model derivation for representing the coupled translations and rotational motion over a friction-less table that is required for control design step in the sequel, b) a nonlinear dynamic inversion based controller framework, while also presenting a low-level optimal actuator allocation logic, specifically designed for the proposed platform enabling an accurate actuation, explicitly accounting thruster physical limitations such as magnitude constraints and minimum on time, c) Experimental validation of autonomous motion achieving a fast translation to the desired tracking behaviour. As it will be presented, the nonlinear dynamic inversion scheme is a state feedback based nonlinear control approach that accounts for the nonlinearity of friction-less floating platform dynamic model explicitly while eliminating the requirement for tedious gain scheduling. Moreover, the closed-form state feedback solution makes the computational process extremely fast for real-time implementation, specifically for space applications, where the processing capability [22] is very limited.

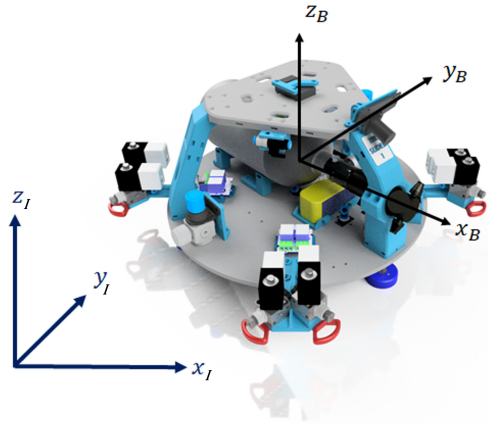


Fig. 1: Schematic of the designed PFP platform with the body-fixed and inertial coordinate frames represented as $\mathbb{B} = \{x_B, y_B, z_B\}$ and $\mathbb{I} = \{x_I, y_I, z_I\}$, respectively.

The rest of the article is structured as follows. In Section II, the mathematical model of the floating platform is derived, while a brief description of the nonlinear dynamic inversion based control formulation is presented in Section III. The overall efficiency of the proposed control scheme is validated with experimental results in Section IV, while the article is concluded with future directions in Section V.

II. MODELLING OF PLANAR FLOATING SATELLITE PLATFORM

The proposed design of the considered Planar Floating satellite Platform (PFP), also called Slider, is depicted in Fig. 1. The PFP is designed to emulate the zero-gravity motion of the spacecraft over a flat friction-less table and it is supported with three air-bearings, attached at its bottom deck. The functional surface of each air-bearing is porous in nature, through which compressed air is uniformly released that eventually creates an air cushion. This air-cushion, on a relatively flat surface, provides a friction-less translation and rotational motion that emulates zero-gravity conditions within the plane of the flat surface. to control the mobility of the platform over a flat surface, it is equipped with eight thrusters that are capable of providing the required force and torque. The PFP is also equipped with an air tank that stores compressed air for the air-bearings and thrusters. Upon the activation of a desired thruster, compressed air from the air tank is released that generates a constant magnitude of thrust in the opposite direction of the air flow and hence, it is assumed that each thruster is operated with a single operation point, i.e., operated in a binary ON-OFF mode. The platform is designed in a compact and symmetrical manner so as to distribute platform's weight equally on the air bearing as well as maintaining the center of gravity close to center of the platform. More insights on the platform's design are documented in [14].

In order to describe the equations of motion of PFP, two coordinate frames, namely the body-fixed frame ($\mathbb{B} =$

$\{x_B, y_B, z_B\}$) and the inertial frame ($\mathbb{I} = \{x_I, y_I, z_I\}$), are considered as shown in Fig. 1). The body-fixed frame is attached to the PFP's Centre of Mass (CoM), while an inertial frame of reference is assumed to be attached at the centre of the flat table with its x_I , y_I and z_I axis to be directed along the tables length, width and height, respectively.

The position of the PFP, denoted as $x, y \in \mathbb{R}$, is described with respect to the inertial frame $\mathbb{I}$. Since, various sensors and actuators are attached to the PFP's body, it is preferred to define its velocity in the body frame $\mathbb{B}$. Let $v_x, v_y \in \mathbb{R}$ denote the velocity components of the PFP, expressed along the x_B and y_B directions respectively. The PFP's motion along the z_B component is ignored as the air cushion created by using air bearings is fairly uniform and the motion of PFP is confined along the operational surface of the table. The translation kinematic equations of motion are described as:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = R_B^I \begin{bmatrix} v_x \\ v_y \end{bmatrix} \quad (1)$$

where, $R_B^I \in \mathbb{R}^{2 \times 2}$ represents the rotation matrix between body and inertial frames. Since, PFP can rotate only about its z_B axis, the rotational velocity is denoted as ω_z , directed along z_B . Based on the fundamental Newton's law of motion [23], the translation dynamics of the PFP are formulated as:

$$\begin{bmatrix} \dot{v}_x \\ \dot{v}_y \end{bmatrix} = \begin{bmatrix} \omega_z v_y + a_x \\ -\omega_z v_x + a_y \end{bmatrix} \quad (2)$$

where a_x, a_y denote the control accelerations along the x_B and y_B directions, while the translation dynamics incorporate the coriolis effects ($\omega_z v_x, \omega_z v_y$) due to its rotational motion. The formulation of the rotational motion of the PFP is based on the conservation of the angular momentum [23] as:

$$\begin{bmatrix} \dot{\theta} \\ \dot{\omega}_z \end{bmatrix} = \begin{bmatrix} \omega_z \\ \frac{\tau_z}{I_{zz}} \end{bmatrix} \quad (3)$$

where $\theta \in \mathbb{R}$ represents the orientation of the PFP, that is the heading angle between the x_I and x_B axis, $\tau_z \in \mathbb{R}$ denotes the applied torque along the z_B direction and $I_{zz} \in \mathbb{R}$ indicates the principal moment of inertia along the z_B direction. In the presented design, the PFP platform has been designed in a balanced manner, such that the off-diagonal components of the moment of inertia matrix are negligible. Combining (5)-(3), the dynamical equations of motion in a compact form is represented as:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{v}_x \\ \dot{v}_y \\ \dot{\omega}_z \end{bmatrix} = \begin{bmatrix} v_x \cos \theta - v_y \sin \theta \\ v_x \sin \theta + v_y \cos \theta \\ \omega_z \\ \omega_z v_y + \frac{1}{m} f_x \\ -\omega_z v_x + \frac{1}{m} f_y \\ \frac{\tau_z}{I_{zz}} \end{bmatrix} \quad (4)$$

where, f_x, f_y represents the force along the respective direction in body frame of the PFP.

The platform is designed such that all the eight thrusters are aligned in the $x_B - y_B$ plane, so as to provide translation on the table surface and the rotational motion about its z_B axis. The forces and torque components, i.e., control

actions, are manifested by an actuation unit equipped with the assembly of eight pneumatic thrusters attached to the PFP platform. These thrusters are distributed into four brackets and placed in a specific alignment. Let us consider that the resulting control accelerations ($a = [a_x, a_y, a_z]^\top$) and torques ($\tau = [\tau_x, \tau_y, \tau_z]^\top$), generated by the thruster assembly and expressed in the frame $\mathbb{B}$, are defined as:

$$a = \frac{1}{m} \Gamma T, \quad \tau = L T \quad (5)$$

where m denotes the mass of PFP. The force configuration matrix $\Gamma \in \mathbb{R}^{3 \times 8}$ expressed as $\Gamma(:, l) = [\cos \beta_l \sin \beta_l \ 0]^\top \forall l = 1, \dots, 8$. Here $\beta_l \in \mathbb{R}$ represents the alignment of l^{th} thruster, which is essentially the direction of thrust force with respect to x_B and $T = [T_1, T_2, T_3, T_4, T_5, T_6, T_7, T_8]^\top$ is the total thrust force vector. Moreover, $T_l \in \{0, T_h^{max}\} \forall l = 1, \dots, 8$, where T_h^{max} is the maximum thrust, implies that each thruster is operated in a binary ON-OFF mode. Similarly, the torque configuration matrix $L \in \mathbb{R}^{3 \times 8}$ is expressed as $L(:, l) = \tilde{r}_l \Gamma(:, l)$ with $l = 1, \dots, 8$, $r_l \in \mathbb{R}^3 = [r_l^x, r_l^y, r_l^z]^\top$ is the position of each i^{th} thruster, expressed in frame $\mathbb{B}$, and $\tilde{r}_l$ is its skew-symmetric matrix. Since, the platform design is made in such a way that all the eight thrusters are aligned in the $x_B - y_B$ plane, forces and torques are produced only for planar motion of PFP. This facilitate further simplification by implying $r_l^z = 0, a_z = 0$ and $\tau_x = \tau_y = 0$. Hence, (5) can be reduced to consider only the forces in 2-D $x_B - y_B$ plane and rotational torque along the z_B direction. In compact mathematical notation (5) is rewritten as:

$$\begin{bmatrix} a_x \\ a_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} \frac{1}{m} \sum_{l=1}^8 T_l \cos \beta_l \\ \frac{1}{m} \sum_{l=1}^8 T_l \sin \beta_l \\ \sum_{l=1}^8 (T_l r_l^x \sin \beta_l - T_l r_l^y \cos \beta_l) \end{bmatrix} \quad (6)$$

III. NONLINEAR DYNAMIC INVERSION BASED CONTROL DESIGN

In order to establish the autonomous maneuvering capability of the PFP by imitating typical satellite motion, (such as representative orbital motion through a circular trajectory or performing docking with a target satellite at specific location), a nonlinear dynamic inversion based command tracking control design is presented in this Section. With the help of compact mathematical notation, the equation of motion of PFP in (4) can be represented in control affine form as:

$$\dot{X} = f(X) + g(X)U \quad (7a)$$

$$Y = h(X) \quad (7b)$$

where $X \in \mathbb{R}^6 = [x, y, \theta, v_x, v_y, \omega_z]^\top$ denote the states of the PFP, $f(X) : \mathbb{R}^6 \rightarrow \mathbb{R}^6$, $g(X), h(X) : \mathbb{R}^6 \rightarrow \mathbb{R}^3$ are the nonlinear functions of state, $U \in \mathbb{R}^3 = [f_x \ f_y \ \tau_z]^\top$ represents the control input, and $Y = [x \ y \ \theta]^\top \in \mathbb{R}^3$ represents the output. The objective is to design a control command

that can ensure pose (position and orientation collectively) of PFP for all time closely follows a desired trajectory denoted by $Y^d(t) = [x^d(t) \ y^d(t) \ \theta^d(t)]^T$. The error in tracking performance is defined as $e(t) = Y(t) - Y^d(t)$. Control command is obtained by enforcing the following stable error dynamics presented as:

$$\ddot{e} + K_1\dot{e} + K_2e = 0 \quad (8)$$

where, $K_1, K_2 > 0 \in R^{3 \times 3}$ are positive definite matrices, considered to be the control design parameters. From the definition of the tracking error e , it's time derivatives are represented as:

$$\dot{e} = \dot{Y} - \dot{Y}^d, \ddot{e} = \ddot{Y} - \ddot{Y}^d \quad (9)$$

Using the equation of motion in (4), expressions of $\dot{Y}$ and $\ddot{Y}$ are obtained as:

$$\dot{Y} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} v_x \cos \theta - v_y \sin \theta \\ v_x \sin \theta + v_y \cos \theta \\ \omega_z \end{bmatrix} \quad (10a)$$

$$\ddot{Y} = G(\theta, \Gamma)U \quad (10b)$$

$$G(\theta, \Gamma) = \begin{bmatrix} \frac{1}{m} \cos(\theta) & -\frac{1}{m} \sin(\theta) & 0 \\ \frac{1}{m} \sin(\theta) & \frac{1}{m} \cos(\theta) & 0 \\ 0 & 0 & \frac{1}{I_{zz}} \end{bmatrix} \quad (10c)$$

where, $\Gamma = [m, I_{zz}]^T$ represents the parameter vector. Substituting the expressions from (10) and (9) into (8), the control command in state feedback form is obtained as:

$$U = G(\theta, \Gamma)^{-1} \left(\ddot{Y}^d - K_1\dot{e} - K_2e \right) \quad (11)$$

Note that, the parameter vector being function of mass and moment of inertia of PFP Γ is constant. Moreover, it is evident that $G(\theta, \Gamma)^{-1}$ exists and it is non-singular for all variation of θ . However, the nonlinear dynamic inversion based control mechanism generates the required force and torque command as a continuous function of time. In order to realize the control command using the thruster based actuation unit, operating in On-off mode, a thruster selection logic is needed, which is presented next.

Thruster Selection Logic

The thruster selection logic distributes the control command into a set of thrusters, which are required to be activated in order to achieve the desired commanded force and torque inputs. In general to address this issue, pseudo-inverse based solutions approach are adopted for addressing attitude control problems (autopilot design) appears in typical space mission. In this article, the thruster selection logic has been formulated as an equivalent optimization problem described as follows:

$$\text{Min} J = T^T R_T T \quad (12a)$$

$$\text{such that } AT = U_k \quad (12b)$$

$$T_{ub} \geq x \geq T_{lb} \quad (12c)$$

where, A denotes the thrust allocation matrix given by

$$A(:, l) = \begin{bmatrix} \cos \beta_l \\ \sin \beta_l \\ (r_{T_l}^y \cos \beta_l - r_{T_l}^x \sin \beta_l) \end{bmatrix} \quad \forall l = 1 \dots 8$$

The solution of the above optimization problem provides the optimal magnitude of each thruster (T_k^*) that collectively mitigates the necessary force and torque demand while minimizes the actuation effort and ensuring the physical limitations of each thruster. Moreover, the magnitude of each thruster T_k^* obtained based on thruster selection logic can be any value between T_{lb} and T_{ub} limits, which enables to incorporate the actuators physical limitations as well. Note that the magnitude of each thruster, obtained based on thruster selection logic, considers any value between its limits, i.e. continuous in magnitude. However, in practice, these thrusters operate in an ON-OFF mode, i.e. either provide a full actuation with a maximum thrust or operate in no thrust mode. In order to address this, the Pulse Width Modulation (PWM) technique is incorporated for each thruster. The PWM provides the required pulse on time for each thruster. If t_{ON}^l is found to be greater than Δt_c , the minimum on-time for effective activation, then only the l^{th} thruster is activated; otherwise, the command is ignored. The thruster activation is made as follows:

$$T_l = \begin{cases} 1, & \text{if } 0 \leq t \leq t_{ON}^l \text{ and } t_{ON}^l \geq \Delta t_c \\ 0, & \text{otherwise} \end{cases}$$

In this way the thrusters' physical limitation of minimum on time is accounted in the framework.

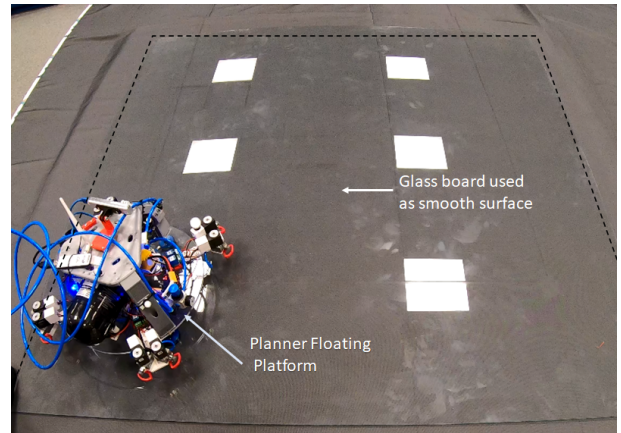


Fig. 2: Experimental testing facility with a 0.7×0.5 m glass covered flat surface, floating platform (Slider) at Luleå University of Technology.

IV. EXPERIMENTAL RESULTS

In order to demonstrate the effectiveness of autonomous manoeuvrability of the PFP platform, extensive experimental studies have been carried out by levitating it over the flat glass table presented in Fig. 2. In this section, different types of controlled motion, which closely resemble robotic space missions, have been emulated with the PFP platform. The

platform is equipped with the Intel RealSense tracking camera T265¹ to provide the real-time state feedback necessary for control algorithms. The T265 has a built-in IMU unit along with a fisheye stereo vision camera, which captures the surrounding environment. Its sensor fusion technology consisting of image processing, feature detection and visual odometry provides fairly accurate localization and velocity (translational as well as rotational) of the PFP platform. The experimental validation is performed by using the dynamic inversion based control design approach incorporated with proposed actuation logic.

A. Case-I: Point-to-point maneuver

In the first case, a point-to-point maneuver for the PFP is aimed. Such types of motion planning, are analogous to a docking scenario, where a spacecraft needs to be guided towards a docking port, while ensuring its terminal position as well as orientation. Hence, a set point tracking should be the priority focus for the utilized control scheme. In the current simulated scenario, initially the PFP is resting, while its initial state is denoted as X_0 . The objective is to translate the PFP to a desired goal position $(x_f, y_f) = (0.4, 0.2)$ m, while ensuring its terminal orientation is specified as $\theta_f = 0^\circ$. The controller gains are considered as $K_1 = I_{3 \times 3}$, $K_2 = 1.5 I_{3 \times 3}$. The set of parameters, utilized for the simulation, is specified in the Table I, while the obtained closed loop trajectory response of the PFP is presented in Fig. 3.

TABLE I: Utilized Experimental Parameters

Parameters	Symbol	Value
Control time step	Δt	0.1 s
Mass of PFP	m	4.27 kg
Moment of Inertia	I_{zz}	0.1092 kg · m ²
Thrust Bound	$T_h^{\max}, T_h^{\min}$	0.7 N, 0 N
Minimum on time	$t_{on}^{\min}$	0.05 s

The variation of PFP's position and orientation with time is presented in Fig. 4. However, from the obtained results, it is evident that the point to point manoeuvre ensures all the terminal constraints. The variation of the thruster based actuation profiles are presented in Fig. 5-6. Initially, the thruster actuation commands are present that enables the correction of the trajectory. Once the PFP reaches its destination point, ideally, thruster ignition should have been stopped. However, due to the presence of slope on the floor over which the friction-less table is placed, an unwanted dragging force acts on the PFP as an external disturbance. The closed-loop control action counteracts the disturbance and stabilizes the PFP over its target point.

B. Case-II: Trajectory tracking maneuver

In the second case, trajectory tracking manoeuvres are illustrated. These types of motion can typically be considered

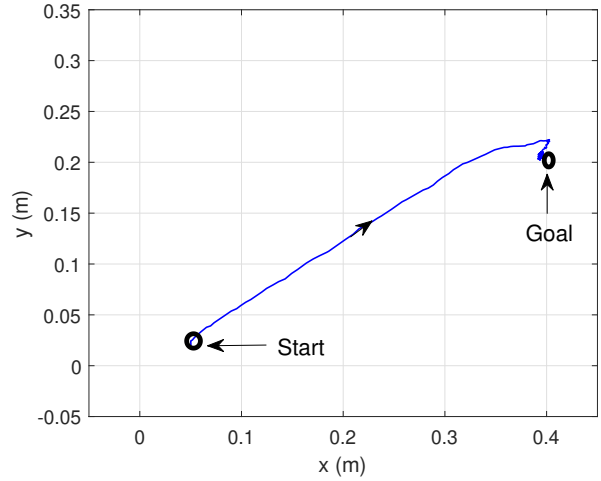


Fig. 3: PFP's response during a point to point manoeuvre

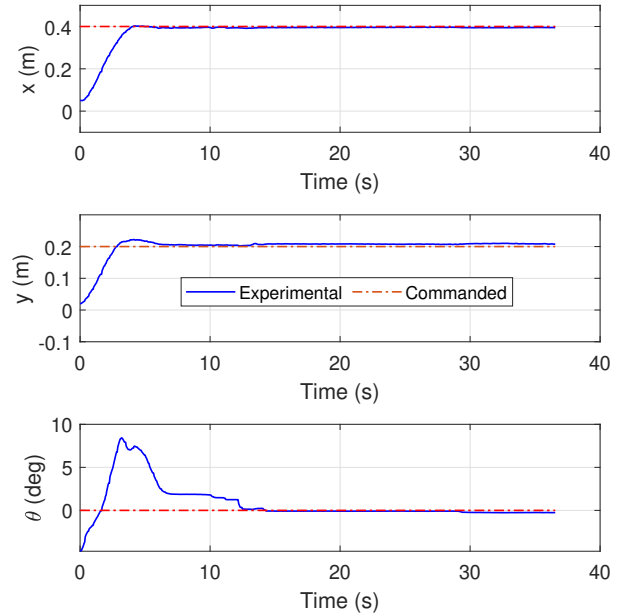


Fig. 4: Time response of PFP's states and corresponding control signals during a point-to-point maneuver

as the foundation of various complex robotic space emulation scenarios, where multiple robotic platforms are involved in demonstrating command tracking leader-follower architecture or imitating a space debris removal mission. In such a case, a robotic module needs to follow a circular or an elliptic trajectory representing satellite orbital motion. In the present experimental setup, thus a circular path following approach is demonstrated, where the PFP needs to follow a circular trajectory described as:

$$\begin{bmatrix} x(t) \\ y(t) \\ \theta(t) \end{bmatrix} = \begin{bmatrix} 0.2 \cos(\omega_d t) \\ 0.2 \sin(\omega_d t) \\ \frac{\pi}{6} \sin(\omega_d t) \end{bmatrix}$$

¹<https://www.intelrealsense.com/tracking-camera-t265/>

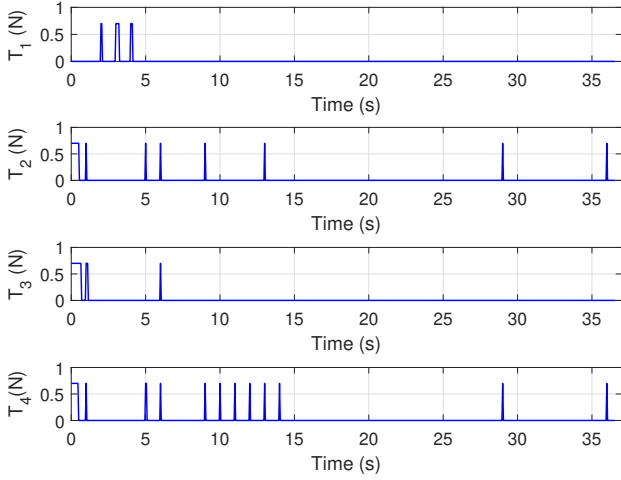


Fig. 5: Thruster $T_1 - T_4$ actuation during a point-to-point maneuver

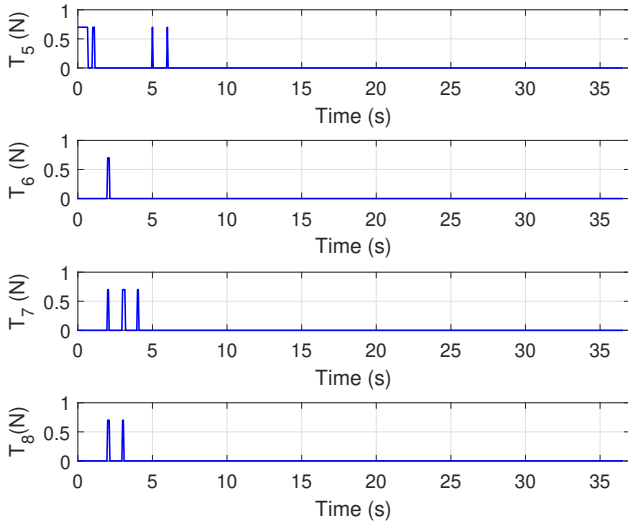


Fig. 6: Thruster $T_5 - T_8$ actuation during a point-to-point maneuver

The obtained PFP trajectory is presented in Fig. 7, where it can be observed that it is having an initial offset from the desired trajectory, and in the sequel it converges and the PFP is able to track the circular path.

Variation of PFP's position (x and y) and orientation θ profiles for circular path tracking are shown in Fig. 8. Since, the initial position of the PFP is not placed exactly over the circle, initial variations are a bit different from a perfect sinusoidal motion. However, the control action brings the PFP quickly on circular trajectory and it follows the sinusoidal variations as specified. It is evident from the trajectories that the output tracking performance which could be a major advance for real-time space applications. Since,

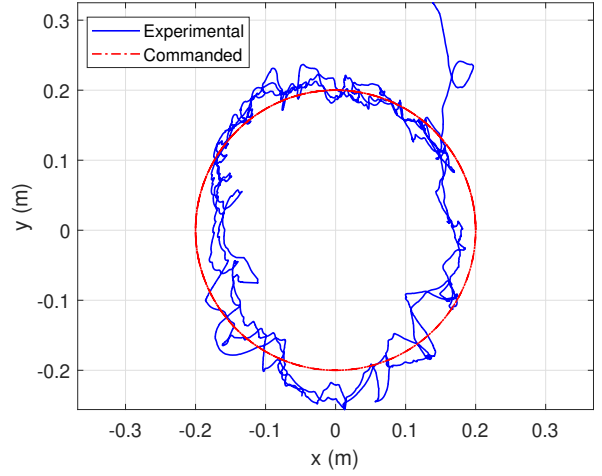


Fig. 7: PFP's response during a circular tracking manoeuvre

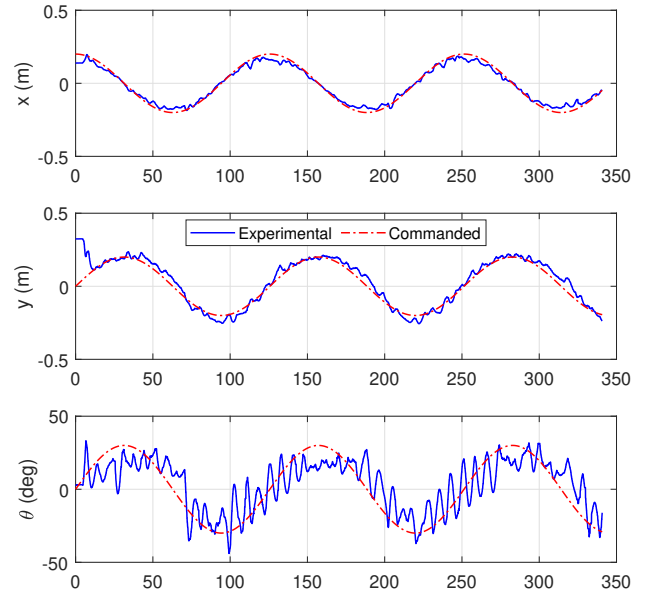


Fig. 8: Time response of PFP's position and orientation during a tracking maneuver

the present experimental setup considered a smooth glass plate placed over the floor, due to presence of the slope associated with the floor, a constant dragging effect along the x direction persist throughout the experimental studies. Hence, a minor deviation is evident for the circular trajectory tracking. However, it establishes the robust behaviour of the proposed control framework.

V. CONCLUSIONS

A brief design description of a friction-less floating robotic test-bed facility for hardware-in-loop experimental study of satellite motion with a nonlinear mathematical model describing its coupled dynamic motion is presented. The

platform provides competence to replicate and test complicated autonomous manoeuvres for the state of the art space technology on the ground with close proximity to a virtual space environment. A nonlinear dynamic inversion based state feedback control design approach augmented with an optimal thruster allocation logic is proposed for controlling the robotic platform. Experimental validation with supported results corroborates that the proposed controller can comprehensively address various motion planning scenarios while accounting for non-linearity of the coupled dynamic and sufficient robustness to account for various unmodelled disturbance effects.

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